

CUSTOMER CHURN & REVENUE RETENTION INTELLIGENCE ENGINE

**Diagnostic analytics and behavioural segmentation engine for customer
churn mitigation and revenue retention in digital commerce.**

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NOTATIONS AND NOMENCLATURE

Term / Notation	Technical & Business Definition
Churn	A binary behavioural indicator identifying accounts that have ceased purchasing within a specified evaluation window.
Tenure	Duration of customer relationship measured in continuous or discrete months.
Recency	Elapsed time in days since a customer's most recent transaction on the platform.
Frequency	Total number of completed orders or transactions executed during the evaluation window.
Monetary	Financial transaction volume or gross monetary value generated by an individual customer.
CSAT	Customer Satisfaction Score evaluated on an ordinal integer scale from 1 to 5.
RFM Segmentation	A quantitative behavioural framework that scores customers across Recency (days since last purchase), Frequency (total orders), and Monetary value (total spend/cashback).
CTE	Common Table Expression; Relational SQL techniques (such as NTILE and DENSE_RANK) employed to perform multi-stage analytical ranking and segmentation.
DAX	Data Analysis Expressions; A functional formula language in Power BI used to dynamically compute measures such as Churn Rate, Retention Rate, and Complaint Risk Multipliers across interactive filters.
Lifecycle Cohort Retention	The longitudinal measurement of customer survival rates grouped by tenure milestones (e.g., 0–3 months, 4–6 months) across customer segments.

1. INTRODUCTION

1.1 Project Overview

In modern digital commerce and retail ecosystems, sustainable business growth relies heavily on retaining existing customers rather than constantly acquiring new ones. Customer acquisition is widely recognized to be significantly more expensive than retention. Customer attrition—commonly termed "churn" represents the cessation of transactions or engagement by an existing buyer. This phenomenon directly reduces top-line revenue, diminishes customer lifetime value, and inflates customer acquisition burn rates.

The project titled "Customer Churn & Revenue Retention Intelligence Engine" is an end-to-end analytical solution developed to investigate, diagnose, and quantify customer attrition. By transforming raw, multi-attribute transactional data into structured business intelligence, the system enables commercial teams to transition from reactive loss management to proactive, data-driven retention strategies.

1.2 Field of the Project

This project lies in the specialized domain of Data Analytics and Business Intelligence (BI), sitting at the intersection of Applied Data Science, Database Systems, and Commercial Decision Support Systems.

1.3 Technologies Used

The engine utilizes a modern, core data analytics technology stack:

1. Python (Pandas, NumPy, Matplotlib, Seaborn): Used for automated data ingestion, audit checks, median-based missing value imputation, categorical standardisation, and statistical exploratory data analysis (EDA).
2. Structured Query Language (SQL) & Relational Databases (MySQL / SQLite): Used for analytical data mart creation, constraint enforcement, and query optimization.
3. Microsoft Power BI: Used for interactive visual analytics, dimensional data modelling (Star Schema), and calculation of operational business KPIs.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

In digital commerce, customer attrition frequently occurs as "silent churn" users gradually reduce their purchase frequency, disengage from mobile applications, or switch to competitors without providing explicit cancellation feedback. Organizations often store transactional, operational, and customer service data in disconnected silos. Without an integrated analytical data mart, business leaders face three critical operational bottlenecks:

1. **Lack of Lifecycle Visibility:** Commercial teams cannot readily identify at which tenure milestone customers are most vulnerable to attrition.
2. **Unquantified Revenue Exposure:** Organisations fail to measure the monetary volume tied to disengaging accounts, making it difficult to calculate the financial impact of customer dissatisfaction.
3. **Inefficient Retention Spending:** Without granular customer segmentation, retention marketing budgets and promotional discount codes are distributed indiscriminately, rather than targeted toward high-value accounts at immediate risk of churning.

2.2 Objectives of the Project

To resolve these industry challenges, this minor project achieves the following concrete technical and analytical objectives:

1. **Data Cleaning and Standardisation:** Implement an automated Python pipeline to handle missing values via median imputation, eliminate categorical spelling inconsistencies, enforce proper data types, and treat fulfilment distance outliers across 5,630 customer records.
2. **Relational Database Mart Construction:** Model and populate a relational database schema (MySQL / SQL) with table indexing to support diagnostic queries using Common Table Expressions (CTEs) and window functions (NTILE, DENSE_RANK).
3. **Diagnostic Exploratory Data Analysis (EDA):** Identify primary behavioural and demographic drivers of churn, specifically measuring the impact of product categories, delivery distance, and customer complaints.
4. **Quantitative RFM Customer Segmentation:** Construct an empirical Recency, Frequency, and Monetary (RFM) engine to classify customers into five distinct strategic tiers (*Champions, Loyal Customers, Potential Loyalists, At Risk, and Lost/Hibernating*).
5. **Tenure Lifecycle Cohort Retention Modelling:** Construct multi-dimensional cohort retention matrices to map customer survival rates across tenure milestones and geographic city tiers.
6. **Executive Business Intelligence Dashboarding:** Build an interactive 3-page Power BI dashboard featuring dynamic DAX measures, cross-filtering slicers, and customer-level drill through grids.

3. FEASIBILITY STUDY

The feasibility study serves as the initial phase in the software engineering lifecycle, evaluating whether the proposed solution satisfies technical, economic, and operational requirements while establishing project necessity.

3.1 Need and Significance of the Project

Customer acquisition in digital commerce costs considerably more than customer retention. Most organisations lack unified diagnostic systems to monitor customer lifecycles, leading to silent attrition and unquantified revenue loss. This project addresses an essential industry requirement by developing an empirical analytics engine that consolidates data wrangling, relational querying, behavioural segmentation, and interactive visual reporting. The significance of this work lies in replacing intuition with data-driven metrics, enabling commercial teams to diagnose churn patterns and safeguard revenue before customer loss becomes irreversible.

3.2 Feasibility Analysis

- i. **Technical Feasibility:** The architecture relies on standard, reliable technologies: Python (Pandas, NumPy) for data processing, MySQL/SQLite for relational querying, and Microsoft Power BI for dashboarding. The entire pipeline executes locally on commodity hardware (8 GB RAM, standard multi-core processor) with sub-minute execution times, requiring no cloud infrastructure.
- ii. **Economic Feasibility:** The software stack comprises open-source tools and free academic desktop applications, resulting in zero software procurement costs. In an enterprise deployment, the engine provides substantial financial value by identifying high-risk segments accounting for over \$400,000 in spend exposure, delivering high return on investment.
- iii. **Operational Feasibility:** The final deliverable is an intuitive Power BI dashboard with simple graphical slicers. Non-technical marketing and retention executives can readily filter cohorts and review customer risk profiles without writing code, ensuring seamless organisational adoption.

4. METHODOLOGY / PLANNING OF WORK

The project adopts a structured, phase-wise data engineering and business intelligence lifecycle. The methodology bridges raw transactional records with executive decision support through five sequential operational stages.

Raw Data Ingestion & Audit → Automated Data Cleaning → Relational Database (SQL)
→ RFM Scoring & Cohort Retention → Interactive Power BI Dashboard.

4.1 Step-by-Step Project Development Workflow

Step 1: Data Acquisition, Auditing, and Data Dictionary.

The initial stage involves importing the raw e-commerce dataset (5,630 records, 20 attributes). A preliminary data audit is executed in Python to inspect data types, detect duplicate records, calculate missing value percentages, and document a structured data dictionary mapping technical fields to business definitions.

Step 2: Automated Data Cleaning and Preprocessing.

Data cleaning is conducted using Pandas to ensure analytical validity:

1. **Categorical Standardisation:** Harmonised inconsistent text labels (e.g., standardising Phone to Mobile Phone, and consolidating payment codes like COD into Cash on Delivery).
2. **Distribution-Preserving Imputation:** Imputed missing values across critical numerical attributes (Tenure, DaySinceLastOrder, OrderCount, WarehouseToHome) using median values rather than means to prevent distortion from skewed distributions.
3. **Integrity Constraints and Outlier Capping:** Converted discrete variables to strict integer formats and applied threshold capping to physical delivery distances (WarehouseToHome bounded at 100 km).
4. **Exporting Clean Mart:** Stored the validated dataset as `ecommerce_churn_clean.csv`.

Step 3: Relational Schema Modelling and SQL Analytics

The cleaned dataset is ingested into a relational database engine (MySQL / SQLite) via SQLAlchemy:

1. Constructed table schemas with primary keys and secondary indices on query-intensive columns (Churn, PreferredOrderCat, Tenure).
2. Formulated analytical SQL scripts utilising Common Table Expressions (CTEs), multi-level aggregations, and window functions (DENSE_RANK, NTILE) to evaluate churn across categories, complaints, and delivery distances.

Step 4: Quantitative RFM Segmentation and Cohort Retention Modelling.

Customer behavioural segmentation is engineered using empirical quintile ranking:

1. **Recency (R):** Derived from days elapsed since the customer's last order.
2. **Frequency (F):** Derived from total completed orders.
3. **Monetary (M):** Quantified using the total transaction/cashback volume.
4. **Scoring Mechanism:** Customers are scored from 1 to 5 along each dimension using SQL window functions:
5. `R_Score = NTILE(5) OVER (ORDER BY Recency DESC)`
6. `F_Score = NTILE(5) OVER (ORDER BY Frequency ASC)`
7. `M_Score = NTILE(5) OVER (ORDER BY Monetary ASC)`

8. **Strategic Mapping:** Accounts are categorized into five mutually exclusive tiers: *Champions* ($R \geq 4, F \geq 4, M \geq 4$), *Loyal Customers* ($R \geq 3, F \geq 3$), *Potential Loyalists* ($R \geq 3, F < 3$), *At Risk* ($R \leq 2, F \geq 3$ or $M \geq 3$), and *Lost / Hibernating* ($R \leq 2, F < 3$).
9. **Tenure Lifecycle Cohorts:** Grouped customer lifecycles into five intervals (0–3 months, 4–6 months, 7–12 months, 13–24 months, and 25+ months) to generate retention matrices across geographic city tiers.

Step 5: Power BI Dashboard Engineering and DAX Formulation

The analytical tables are modelled in Microsoft Power BI Desktop:

1. Established relationships linking customer profiles with segmented RFM attributes.
2. Formulated a centralized _Measures table containing 10 core DAX metrics, including Churn Rate %, Retention Rate %, Revenue Lost to Churn, and Complaint Rate %.
3. Built an executive 3-page reporting suite featuring dynamic slicers, heatmap visualisations, and drill-through customer tables.

4.2 Planning of Work and Schedule

Table 1: Phase-Wise Work Breakdown and Implementation Schedule

Phase	Major Tasks & Deliverables	Timeline
Phase I	Problem formulation, dataset selection, and exploratory inspection.	Weeks 1–2
Phase II	Python data cleaning, imputation, and categorical standardisation.	Weeks 3–4
Phase III	Database schema design, SQL data ingestion, and diagnostic querying.	Weeks 5–6
Phase IV	RFM mathematical scoring and multi-dimensional cohort matrix generation.	Weeks 7–8
Phase V	Power BI Star Schema modelling, DAX formulation, and 3-page dashboard build.	Weeks 9–10
Phase VI	Business insight formulation, synopsis preparation, and minor project report.	Weeks 11–12

5. FACILITIES REQUIRED FOR PROPOSED WORK

The development, execution, and validation of the project require the following hardware and software facilities:

5.1 Hardware Requirements

1. Processor: Intel Core i3 / i5 or AMD Ryzen equivalent (minimum dual-core, 2.0 GHz or higher).
2. Random Access Memory (RAM): Minimum 8 GB (16 GB recommended for seamless Power BI in-memory data virtualization).
3. Storage: Minimum 256 GB Solid State Drive (SSD) with at least 5 GB of free space for datasets, databases, and dependencies.
4. Display: Standard monitor supporting a minimum resolution of 1366×768 (1920×1080 Full HD recommended).

5.2 Software Requirements

1. Operating System: Microsoft Windows 10 / 11 (64-bit).
2. Programming Language & Environment: Python 3.10+ executed via JupyterLab / Jupyter Notebook.
3. Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn, SQLAlchemy, and PyMySQL.
4. Database Management System: MySQL Workbench 8.0+ / SQLite 3 for relational data storage and SQL querying.
5. Business Intelligence Software: Microsoft Power BI Desktop (May 2024 or later release) for dashboard engineering and DAX calculations.
6. Version Control: Git and GitHub for source code management.

6. SCOPE OF THE PROJECT

The practical application of the Customer Churn & Revenue Retention Intelligence Engine is designed for deployment within modern digital commerce platforms, direct-to-consumer (D2C) brands, and online retail enterprises. The primary beneficiaries include:

1. **Customer Retention Managers:** Can directly utilise the RFM segmentation engine and customer action table to identify dormant high-spenders (the \$400K "At-Risk" cohort) and deploy automated win-back marketing campaigns.
2. **Customer Support Leadership:** Can prioritize complaint ticket resolution for high-value accounts, neutralizing the 2.9× churn surge associated with unresolved customer dissatisfaction.
3. **Category Merchandisers:** Can isolate underperforming product lines—such as the Mobile Phone category experiencing an elevated 27.4% churn rate—and introduce bundled warranties, accessories, or loyalty incentives to encourage repeat purchases.
4. **Executive Leadership:** Can leverage the interactive Power BI dashboard to monitor real-time retention rates, evaluate geographic fulfilment risks across city tiers, and protect enterprise cash flows.

The framework provides an immediately deployable decision support system that transforms raw transaction logs into targeted commercial interventions.

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